INCH-POUND
MIL-PRF-55365/4E
3 July 1997
SUPERSEDING
MIL-C-55365/4D
29 March 1994

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLES CWR06 AND CWR09

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55365.

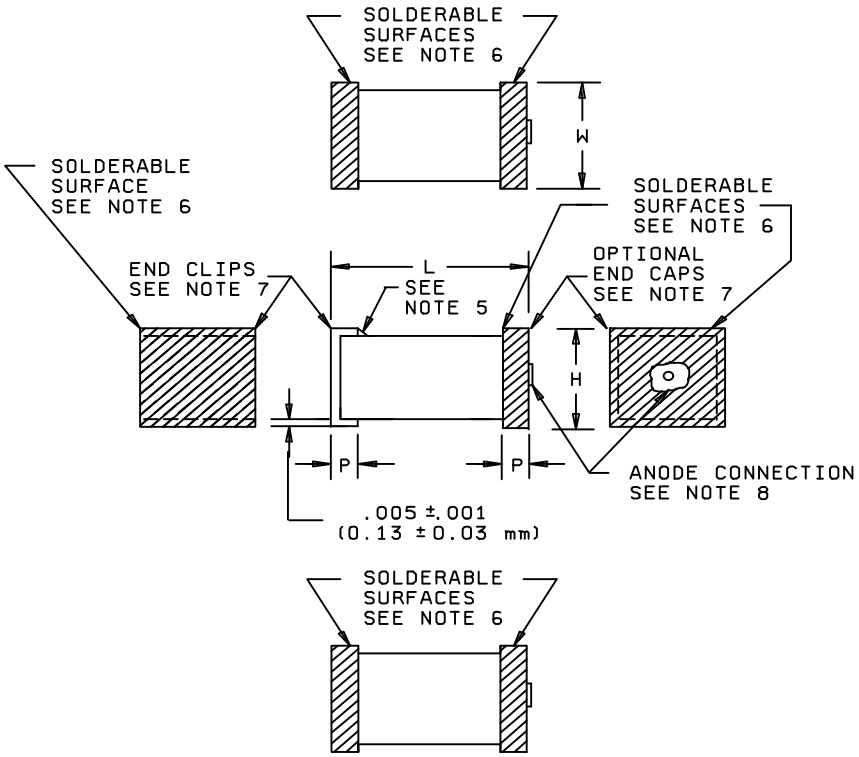

FIGURE 1. Style CWR06 (conformally coated) capacitors.

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Case size	L	W	H	P
	± .015 (0.38)	±.015 (0.38)	±.015 (0.38)	±.005 (0.13)
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. Cathode terminal may be epoxy coated on one surface for polarity identification.
6. Solderable surfaces are only those surfaces designated as such.
7. Anode and cathode terminals may be two-sided through five-sided, and may not be the same on some designs, but will meet all specification requirements.
8. The anode terminal shall be identified by the riser wire connection, which may extend the case size .015 (0.38 mm) maximum. The riser wire weld area is not solderable. The riser wire connection shall have a continuous weld of a minimum of 25 percent of the contact surface.
9. When solder coated terminations are required, add .015 (0.38 mm) to termination dimension tolerances.

FIGURE 1. Style CWR06 (conformally coated) capacitors - Continued.

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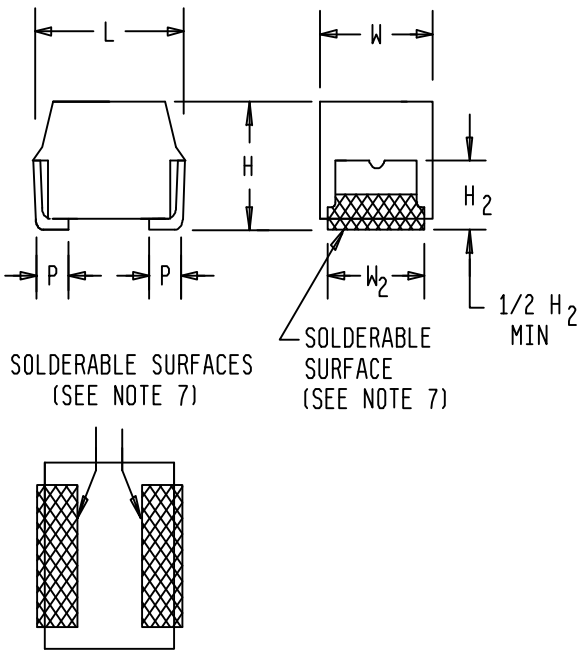

Case size	L	W	H	P	W_2	H_2
	± .015 (0.38)	±.015 (0.38)	±.015 (0.38)	±.005 (0.13)		min
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ±.005 (1.27±.013)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ±.005 (1.27±.013)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ±.005 (1.27±.013)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095+.005,-.010 (2.41+0.13, -0.25)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095+.005,-.010 (2.41+0.13, -0.25)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)	.130±.005 (.330±0.13)	.040 (1.02)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)	.105±.005 (2.67±0.13)	.060 (1.52)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)	.145+.005. -.020 (3.68+0.13, -0.51)	.060 (1.52)

FIGURE 2. Style CWR09 molded chip.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. The anode (+) terminal shall be identified by a marking on the case.
6. The termination width (W_2) shall be maintained to the top of the solderable area.
7. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
8. When solder coated terminations are required, add .015 (0.38 mm) to the termination dimension tolerances.

FIGURE 2. Style CWR09 molded chip - Continued

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Termination finish: In accordance with termination finishes B, C, H, or K of MIL-PRF-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365 (see table I).

Resistance to soldering heat: In accordance with MIL-PRF-55365.

Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Step 1 (+25°C):	DCL	-	See table I.
Cap.	-		Shall not exceed tolerance of table I value.
DF	-		See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
DF	-		See table I.
Step 3 (+25°C):	DCL	-	See table I.
ΔCap.	-		Shall not exceed ±5 percent of initial +25°C measurement.
DF	-		See table I.

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Step 4 (+85°C): DCL - See table I.
ΔCap. - Shall not exceed ±10 percent of initial +25°C measurement.
DF - See table I.

Step 5 (+125°C): DCL - See table I.
ΔCap. - Shall not exceed ±15 percent of initial +25°C measurement.
DF - See table I.

Step 6 (+25°C): DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial +25°C measurement.
DF - See table I.

Surge voltage: In accordance with MIL-PRF-55365.

DCL - See table I.
ΔCap. - Within ±5 percent of initial value.
DF - See table I.

Surge current (optional): This test, when applicable, shall be performed on a 100 percent basis following voltage conditioning of the qualification and group A tests. Capacitors shall be subjected to ten consecutive cycles under the following conditions:

Temperature: Option A: +25°C ±5°C; option B: -55°C ±5°C and +85°C ±5°C.

Applied voltage: Rated dc voltage ±2 percent from a power source having an energy storage bank of 50,000 μf (minimum) across the output terminals.

Charge cycle: 4 seconds ±1 second.

Discharge cycle: 4 seconds ±1 second to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing, and output impedance of the regulated power supply to each test position during the charging cycle shall be 1 ohm ±0.2 ohm.

On completion of the test, the capacitors shall meet the following requirements:

DCL: See table I.

Cap.: Shall meet the initial measured value.

DF: See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap.-Shall not exceed ±5 percent of initial measured value.
DF - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

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10,000 hours:

At 25°C: DCL - 200 percent of initial requirement.
ΔCap.- Shall not exceed ±5 percent of initial measured value.
DF - See table I.

At 85°C: DCL - 200 percent of initial requirement.

At 125°C: DCL - 200 percent of initial requirement.

TABLE I. Styles CWR06 and CWR09 characteristics.

Pin 1/ 2/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C Style CWR06	Max ESR 100 kHz +25°C Style CWR09	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C			
	Volts	µF	µA	µA	µA	%	%	%	Ohms	Ohms	
CWRO-C-225---	4	2.2	1.0	10	12	6	8	8	8.0	8.0	A
CWRO-C-475---	4	4.7	1.0	10	12	6	8	8	8.0	8.0	B
CWRO-C-685---	4	6.8	1.0	10	12	6	8	8	5.5	5.5	C
CWRO-C-106---	4	10.0	1.0	10	12	8	8	10	4.0	4.0	D
CWRO-C-156---	4	15.0	1.0	10	12	8	10	12	3.5	3.5	E
CWRO-C-336---	4	33.0	2.0	20	24	8	10	12	2.2	2.2	F
CWRO-C-686---	4	68.0	3.0	30	36	10	12	12	1.1	1.1	G
CWRO-C-107---	4	100.0	4.0	40	48	10	12	12	.09	.09	H
CWRO-D-155---	6	1.5	1.0	10	12	6	8	8	8.0	8.0	A
CWRO-D-335---	6	3.3	1.0	10	12	6	8	8	8.0	8.0	B
CWRO-D-475---	6	4.7	1.0	10	12	6	8	8	5.5	5.5	C
CWRO-D-685---	6	6.8	1.0	10	12	6	8	8	4.5	4.5	D
CWRO-D-106---	6	10.0	1.0	10	12	8	10	12	3.5	3.5	E
CWRO-D-226---	6	22.0	2.0	20	24	8	10	12	2.2	2.2	F
CWRO-D-476---	6	47.0	3.0	30	36	10	12	12	1.1	1.1	G
CWRO-D-686---	6	68.0	4.0	40	48	10	12	12	.09	.09	H
CWRO-F-105---	10	1.0	1.0	10	12	6	8	8	12.0	10.0	A
CWRO-F-225---	10	2.2	1.0	10	12	6	8	8	8.0	8.0	B
CWRO-F-335---	10	3.3	1.0	10	12	6	8	8	5.5	5.5	C
CWRO-F-475---	10	4.7	1.0	10	12	6	8	8	4.5	4.5	D
CWRO-F-685---	10	6.8	1.0	10	12	6	8	8	3.5	3.5	E
CWRO-F-156---	10	15.0	2.0	20	24	8	8	10	2.5	2.5	F
CWRO-F-336---	10	33.3	3.0	30	36	10	12	12	1.1	1.1	G
CWRO-F-476---	10	47.0	5.0	50	60	10	12	12	.09	.09	H
CWRO-H-684---	15	.68	1.0	10	12	6	8	8	12.0	12.0	A
CWRO-H-155---	15	1.5	1.0	10	12	6	8	8	8.0	8.0	B
CWRO-H-225---	15	2.2	1.0	10	12	6	8	8	5.5	5.5	C
CWRO-H-335---	15	3.3	1.0	10	12	6	8	8	5.0	5.0	D
CWRO-H-475---	15	4.7	1.0	10	12	6	8	8	4.0	4.0	E
CWRO-H-106---	15	10.0	2.0	20	24	6	8	8	2.5	2.5	F
CWRO-H-226---	15	22.0	4.0	40	48	6	8	8	1.1	1.1	G
CWRO-H-336---	15	33.3	5.0	50	60	8	8	10	.09	.09	H
CWRO-J-474---	20	.47	1.0	10	12	8	8	10	16.0	14.0	A
CWRO-J-684---	20	.68	1.0	10	12	6	8	8	14.0	10.0	B

See footnotes at the end of table

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TABLE I. Styles CWR06 and CWR09 characteristics - Continued.

Pin 1/ 2/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C Style CWR06	Max ESR 100 kHz +25°C Style CWR09	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C			
	Volts	µF	µA	µA	µA	%	%	%	Ohms	Ohms	
CWRO-J-105---	20	1.0	1.0	10	12	6	8	8	12.0	12.0	B
CWRO-J-155---	20	1.5	1.0	10	12	6	8	8	6.0	6.0	C
CWRO-J-225---	20	2.2	1.0	10	12	6	8	8	5.0	5.0	D
CWRO-J-335---	20	3.3	1.0	10	12	6	8	8	4.0	4.0	E
CWRO-J-685---	20	6.8	2.0	20	24	6	8	8	2.4	2.4	F
CWRO-J-156---	20	15.0	3.0	30	36	6	8	8	1.1	1.1	G
CWRO-J-226---	20	22.0	4.0	40	48	6	8	8	0.9	0.9	H
CWRO-K-334---	25	.33	1.0	10	12	6	8	8	15.0	15.0	A
CWRO-K-684---	25	.68	1.0	10	12	6	8	8	10.0	7.5	B
CWRO-K-105---	25	1.0	1.0	10	12	6	8	8	6.5	6.5	C
CWRO-K-155---	25	1.5	1.0	10	12	6	8	8	6.5	6.5	D
CWRO-K-225---	25	2.2	1.0	10	12	6	8	8	3.5	3.5	E
CWRO-K-475---	25	4.7	2.0	20	24	6	8	8	2.5	2.5	F
CWRO-K-685---	25	6.8	2.0	20	24	6	8	8	1.2	1.2	G
CWRO-K-106---	25	10.0	3.0	30	36	6	8	8	1.4	1.4	G
CWRO-K-156---	25	15.0	4.0	40	48	6	8	8	1.0	1.0	H
CWRO-M-224---	35	.22	1.0	10	12	6	8	8	24.0	18.0	A
CWRO-M-474---	35	.47	1.0	10	12	6	8	8	17.0	10.0	B
CWRO-M-684---	35	.68	1.0	10	12	6	8	8	10.0	8.0	C
CWRO-M-105---	35	1.0	1.0	10	12	6	8	8	6.5	6.5	D
CWRO-M-155---	35	1.5	1.0	10	12	6	8	8	4.5	4.5	E
CWRO-M-335---	35	3.3	1.0	10	12	6	8	8	2.5	2.5	F
CWRO-M-475---	35	4.7	2.0	20	24	6	8	8	1.5	1.5	G
CWRO-M-685---	35	6.8	3.0	30	36	6	8	8	1.3	1.3	H
CWRO-N-104---	50	.10	1.0	10	12	6	8	8	75.0	22.0	A
CWRO-N-154---	50	.15	1.0	10	12	6	8	8	25.0	17.0	A
CWRO-N-224---	50	.22	1.0	10	12	6	8	8	17.0	14.0	B
CWRO-N-334---	50	.33	1.0	10	12	6	8	8	12.0	12.0	B
CWRO-N-474---	50	.47	1.0	10	12	6	8	8	8.0	8.0	C
CWRO-N-684---	50	.68	1.0	10	12	6	8	8	7.0	7.0	D
CWRO-N-105---	50	1.0	1.0	10	12	6	8	8	6.0	6.0	E
CWRO-N-155---	50	1.5	1.0	10	12	6	8	8	4.0	4.0	F
CWRO-N-225---	50	2.2	2.0	20	24	6	8	8	2.5	2.5	F
CWRO-N-335---	50	3.3	2.0	20	24	6	8	8	2.0	2.0	G
CWRO-N-475---	50	4.7	3.0	30	36	6	8	8	2.0	2.0	H

1/ Complete PIN shall include additional symbols to indicate style, termination finish, capacitance tolerance, product level, and, if applicable, surge current option letter A (+25°C ±5°C) or B (-55°C ±5°C and +85°C ±5°C). If optional surge current is not required, the last "-" shall be deleted.

2/ Styles CWR06 and CWR09 are, with the acquiring agency approval, considered to be interchangeable, provided the size, capacitance, capacitance tolerance, failure rate level, rated voltage, maximum ESR value, and surge current option (if applicable) remain the same.

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Navy - EC
Air Force - 85

Review activities:
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Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
NASA - NA

Preparing activity:
Army - CR

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